



MITTALX ONSTM

INTELLIGENT BIO-TOILET ODOUR NEUTRALISING SYSTEM

ELEVATING USER COMFORT THROUGH CONTINUOUS AIR TREATMENT

MittalX ONSTM is a next-generation odour neutralising system engineered to continuously treat and purify the air in railway Bio Toilets.

Designed for reliability, safety and low maintenance, ONSTM ensures a cleaner, fresher and more comfortable travel experience for passengers.



APPROVED & VALIDATED BY

RDSE

Govt of India
Ministry of Railways

KEY BENEFITS



CONTINUOUS OPERATION

24/7 air treatment for enhanced passenger comfort.



COMPACT FOOTPRINT

Space-saving design ideal for railway applications.



LOW MAINTENANCE, LONG LIFE

Engineered for durability with minimal maintenance requirements.



POWERFUL, ADVANCED AIRFLOW

High airflow efficiency for effective odour neutralisation.





ONS™

INTELLIGENT ODOUR NEUTRALISING SYSTEM

CONTINUOUSLY. QUIETLY. EFFECTIVELY.

HKM ONS™ is an advanced odour neutralising system engineered for railway coaches and other enclosed environments. It continuously reduces odour-causing compounds and improves air quality, creating a cleaner and more comfortable passenger experience.



PROPRIETARY MULTI-STAGE AIR TREATMENT TECHNOLOGY

HKM ONS™ utilises a proprietary multi-stage air treatment process that continuously neutralises odour-causing compounds through multiple complementary technologies. This integrated approach ensures high efficiency, reliability and long service life with minimal maintenance.

HOW IT WORKS



CONTAMINATED AIR

Odour-causing air enters the system.



INTELLIGENT AIR INTAKE

Air is drawn in efficiently and quietly.



PROPRIETARY TREATMENT PROCESS

Multi-stage treatment neutralises a wide range of odour compounds.



CLEAN & FRESH AIR

Clean, odour-free air is released back into the environment.



Designed for continuous operation with low noise, low maintenance and high reliability in demanding railway environments.

KEY FEATURES



Proprietary multi-stage odour neutralisation process



Effectively targets a wide range of odour-causing compounds



Continuous operation for consistent air quality



Compact, robust & vibration-resistant design



Low power consumption & long service life



Minimal maintenance & easy to service

TYPICAL APPLICATIONS



TRAIN LAVATORIES



COMPACT WASTE COMPACTORS



WASTE TRANSFER STATIONS



PUMPING STATIONS



SEWAGE TREATMENT PLANTS



PUBLIC RESTROOMS



RDSO APPROVED

All key products are tested & approved by RDSO for deployment across Indian Railways.



COMPLIANT WITH

- EN 45545-2 Railway fire safety standards
- Category 1, Class A/B of EN 61373



CLEANER AIR. BETTER JOURNEYS.

HKM ONS™ ensures a cleaner, fresher and more comfortable environment for passengers and railway staff.

“ Field evaluation demonstrated a measurable reduction in odour-causing compounds while the ONS system remained operational, validating its suitability for railway passenger environments. ”

– Field Validation Statement



REAL-WORLD VALIDATION

Proven Under Real Railway Operating Conditions

Independent field evaluation demonstrating measurable reduction of odour-causing gases in Indian Railway passenger coaches using HKM ONS™ technology.



HKM ONS™ installed inside an Indian Railway passenger coach during field validation.

PROJECT
Research Designs & Standards Organisation (RDSO)
Indian Railways

APPLICATION
Passenger Coach Lavatory

TECHNOLOGY

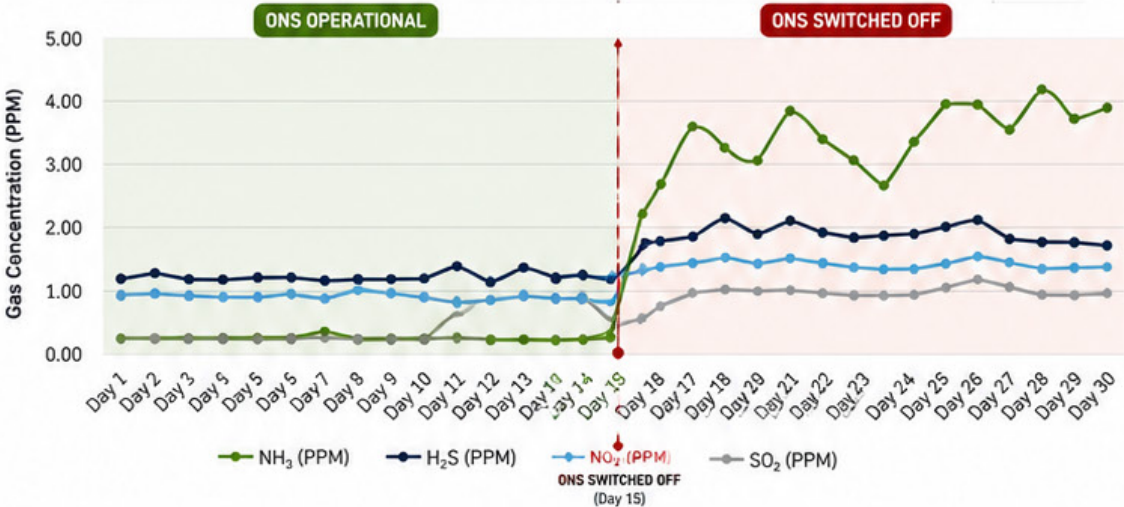
- HKM ONS™
- Activated Carbon Technology
- IoT Gas Monitoring
- Continuous Air Treatment

VALIDATION PERIOD
30-Day Continuous Trial

OBJECTIVE
Measure reduction of odour-causing gases under actual operating conditions.

MEASURED PERFORMANCE

Continuous monitoring of NH₃, H₂S, NO₂ and SO₂ concentrations over a four-week field trial.



OBSERVATION 01
ONS OPERATIONAL

- Stable gas concentration
- Reduced odour levels
- Consistent passenger environment

OBSERVATION 02
ONS SWITCHED OFF

- Immediate increase in gas concentration
- Higher odour intensity
- Clear measurable performance difference

30 DAYS
Continuous Validation

IoT
Real-time Monitoring

RDSO
Field Evaluation

PASSENGER COACH
Live Railway Environment

“Field monitoring demonstrated a measurable reduction in odour-causing gases while the ONS system remained operational, validating its suitability for railway passenger environments.”

PROVEN PERFORMANCE.
Engineered for Modern Railways.

Field validation demonstrated measurable reduction of odour-causing gases under actual railway operating conditions, supporting improved passenger comfort and data-driven maintenance decisions.